DEWDRIVE · TRACEABILITY MATRIX

DewDrive traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

DewDrive traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 12 user requirements, 14 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Water at the design point	DWD-CAL-001, DWD-BLD-001, DWD-STM-001, DWD-OPT-001	Analysis (DWD-CAL-001 cyclic balance), TRL 3; field test at TRL 4	Not met: 0.44 (C3); 0.37 to 0.41 under the sensitivities (C9, C12, C13)
UR-002, UR-001	PR-002	Water in dry nights	DWD-CAL-001, DWD-BLD-001	Analysis (DWD-CAL-001), TRL 3; field test at TRL 4	Met on paper: 0.31 (C4)
UR-008	PR-003	Sun-only regeneration	DWD-CAL-001, DWD-BLD-001	Analysis (collector balance), TRL 3; logged at TRL 4	Met on paper: 105 °C at noon (C2)
UR-008	PR-004	Own electrical loads	DWD-CAL-001, DWD-BLD-001	Analysis (energy budget), TRL 3; measurement at TRL 4	Met on paper: 23.6 against 35 Wh/day; 2.6 days (F1, F2)
UR-001	PR-005	Condenser rise	DWD-CAL-001, DWD-STM-001	Analysis (condenser balance, painted underside D18), TRL 3; logged at TRL 4	At risk: 11.9 K on the correlation (C2); about 14 to 16 K if the fins under the floor shed less heat (independent review)
UR-004	PR-006	Water quality	DWD-CAL-001, DWD-BLD-001	Inspection (materials review), TRL 3; laboratory water test at TRL 4	To verify at TRL 4; materials chosen to meet it
UR-012, UR-004	PR-007	Salt containment	DWD-CAL-001, DWD-BLD-001	Analysis (pore and sump volumes, DWD-CAL-001 E), TRL 3; chamber test at TRL 4	Met on paper: 47 % after one night, 54 % after three with the fan rule; sumps 0.58 L (E2 to E6)
UR-003	PR-008	Two actions per day	DWD-CAL-001, DWD-BLD-001	Analysis (task analysis), TRL 3; co-design sessions at TRL 4	Met by design
UR-007	PR-009	Portable	DWD-CAL-001, DWD-BLD-001	Analysis (mass from model components), TRL 3; trial at TRL 4	Met on paper: 23.4 kg; 36.1 kg with everything inside; 61.1 kg in all (G1, G2)
UR-006	PR-010	Survive the site	DWD-CAL-001, DWD-BLD-001	Analysis (wind load), TRL 3; supplier data and cycling test at TRL 4	To verify at TRL 4: two anchors of 234 N each at 25 m/s, specified 250 N, or 96 kg of ballast (H2 to H4)
UR-010	PR-011	Cost	DWD-CAL-001	Analysis (priced BOM), TRL 3	Over target by $175: $695 (J1)
UR-011	PR-012	Record performance	DWD-CAL-001, DWD-BLD-001	Analysis (storage estimate), TRL 3; demonstration at TRL 4	Met on paper: 8,640 records, 0.41 MB (I1)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-005	PR-013	Protect users	DWD-CAL-001, DWD-BLD-001, DWD-STM-001	Analysis (surface temperatures), TRL 3; logged at TRL 4	Met on paper: glazing outer skin 55 °C, fins 47 °C; 12.8 V DC (D1 to D3)
UR-009	PR-014	Local build and repair	DWD-CAL-001, DWD-BLD-001	Inspection (build plan DWD-BLD-001 and BOM), TRL 3; build at TRL 4	Met by design

Gaps

None.